DATABASES

NoSQL Types Visual Guide

Document, key-value, columnar, graph -- four data models, four superpowers.

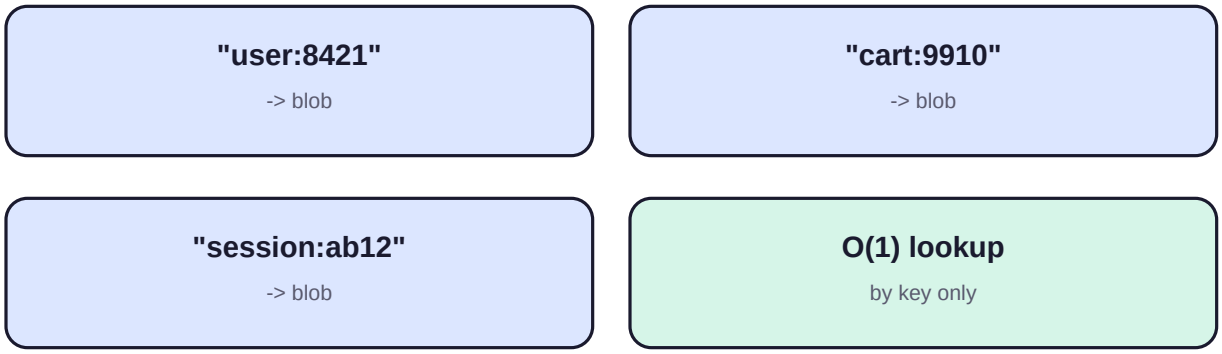
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/5

Key-Value Stores

The simplest model: a giant distributed hash map.



- Strengths: blazing reads/writes, trivial sharding, tiny operational surface.
- Weakness: no queries by value, no joins -- you must know the key.
- Examples: Redis, DynamoDB, Riak. Home of sessions, carts, feature flags, counters.

STEP 2/5

Document Stores

JSON-like documents, indexed and queryable.

```
{ "_id": "u8421", "name": "Ada",
  "tags": ["pro", "beta"],
  "address": { "city": "Lima", "zip": "15001" } }
```

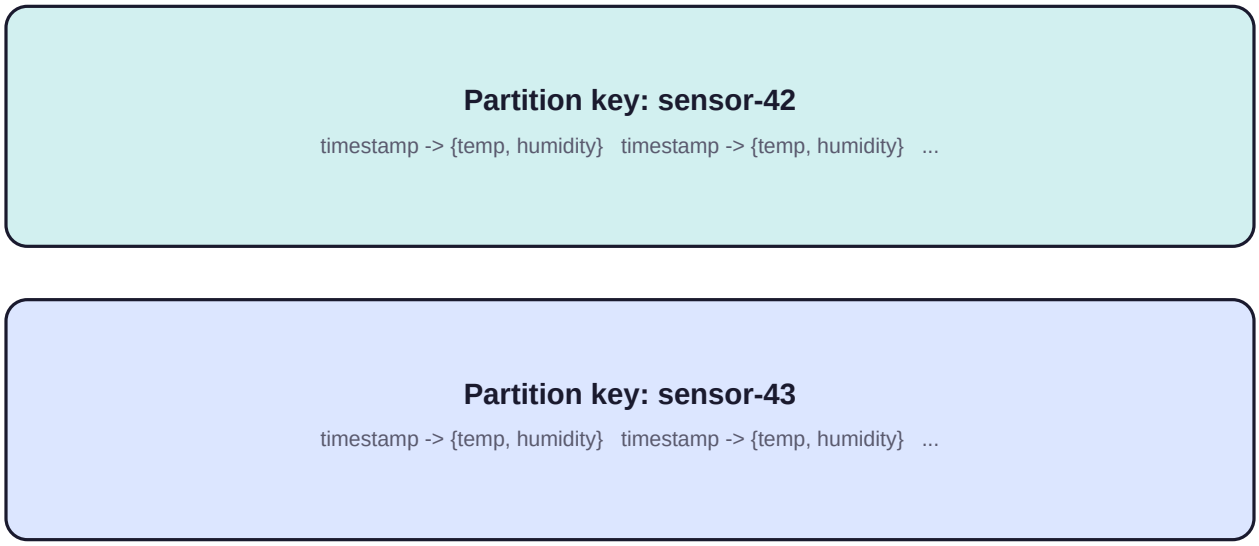
- Strengths: flexible schema, nested data, rich secondary indexes, aggregation pipelines.
- Weakness: joins are awkward, cross-document transactions cost more.
- Examples: MongoDB, Couchbase, Firestore. Home of catalogs, profiles, CMS content.

Embed what you read together; reference what grows unboundedly (comments, events).

STEP 3/5

Columnar / Wide-Column Stores

Rows keyed by partition, columns grouped in families, built for massive scale.

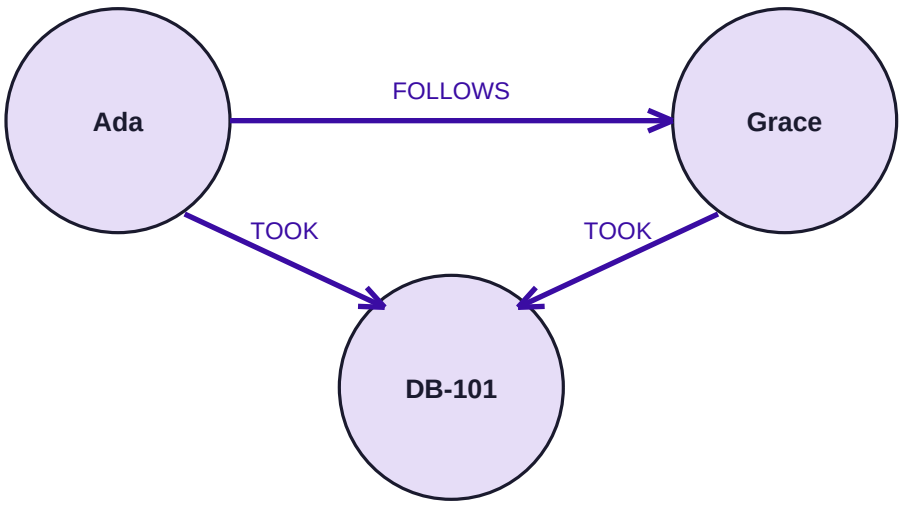


- Strengths: linear write scale-out, cheap time-ordered scans, tunable consistency.
- Weakness: query patterns must be designed up front; ad-hoc joins are painful.
- Examples: Cassandra, ScyllaDB, HBase. Home of time series, IoT, messaging inboxes.

STEP 4/5

Graph Stores

Nodes and edges as first-class citizens.



- Strengths: multi-hop traversals ('friends of friends who took DB-101') in milliseconds.
- Weakness: sharding graphs is hard; poor fit for bulk tabular analytics.
- Examples: Neo4j, Neptune. Home of fraud rings, recommendations, knowledge graphs.

STEP ✓I✓

Key Takeaways

NoSQL Types -- Remember

- Key-value: fastest lookups by key, zero query flexibility -- sessions, caches, counters.
- Document: flexible JSON + real queries -- catalogs, profiles, content.
- Wide-column: massive write scale with pre-designed access patterns -- time series, IoT.
- Graph: relationships as data -- fraud, recommendations, knowledge graphs.
- Choose by access pattern, not hype: model the queries first, then pick the store.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev